NATIONAL JUNIOR COLLEGE
PRELIMINARY EXAMINATION
Higher 1

CANDIDATE NAME	

SUBJECT CLASS		REGISTRATION NUMBER	

CHEMISTRY **8872/02**

Paper 2 Structured Questions and Free Response Questions

Wednesday 5 Sep 2013
2 hours

Additional Materials: Writing Paper
Data Booklet

READ THE INSTRUCTION FIRST

Write your subject class, registration number and name on all the work you hand in.

Do not open this booklet until you are told to do so.

Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working. Do not use paper clips, highlighters, glue or correction fluid.

Section A:
Answer **ALL** questions in the spaces provided on the question paper.

Section B:
Answer **two** questions on separate answer paper.

The number of marks is given in brackets [ ] at the end of each question or part question.

At the end of the examination, fasten all your work together.

You are advised to spend one hour on each section of the paper.

For Examiner's Use		
Section A	1	/13
	2	/9
	3	/7
	4	/11
Section B	6	/20
	7	/20
	8	/20
Total		/80

This document consists of **13** pages and **0** blank page.

For Examiner's Use

Section A: Structured Questions [Total: 40 marks]

Answer **ALL** questions in the space provided.

1 **(a)** Complete the electronic structures of the Cr^{3+} and Mn^{2+} ions.

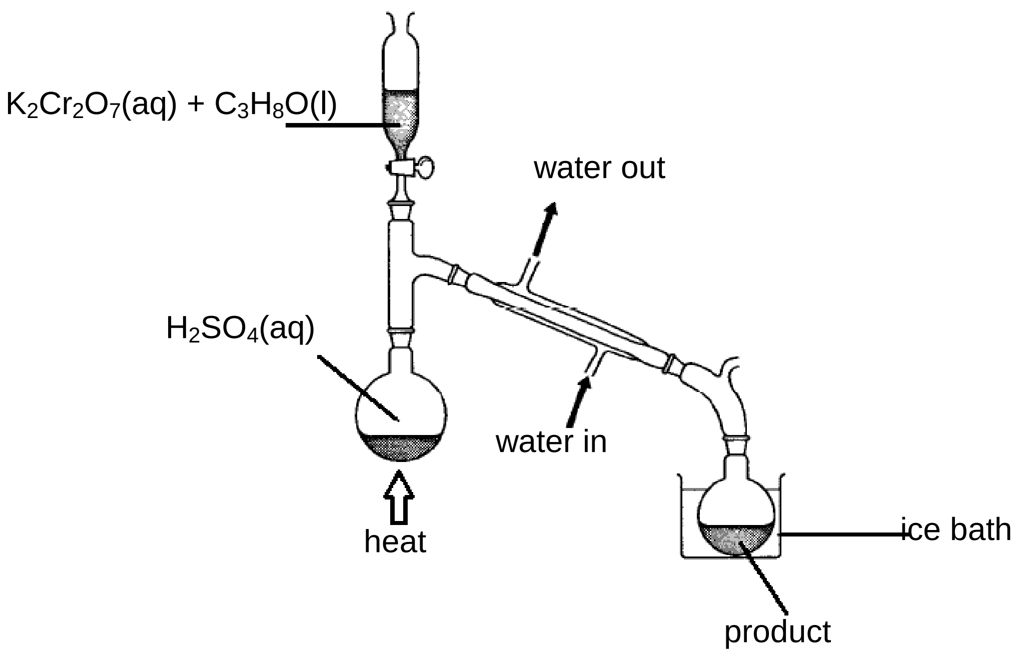
Cr^{3+} $1s^2 2s^2 2p^6$ ....

Mn^{2+} $1s^2 2s^2 2p^6$ ....

[2]

(b) A student was given the following instructions for the oxidation of an alcohol, C_3H_8O.

To 20 cm^3 of water in a round bottom flask, carefully add 5 cm^3 of concentrated sulfuric acid and set up the distillation apparatus as shown below.



Make up a solution containing 47.2 g of potassium dichromate(VI), $K_2Cr_2O_7$, in a 15 cm^3 of water, add 18.0 g of the alcohol, C_3H_8O, and pour this mixture into the dropping funnel.

Boil the acid in the flask. Add the mixture from the dropping funnel at such a rate that the product is slowly collected.

Re-distil the product and collect the fraction that boils between 48^{o}C and 50^{o}C.

For Examiner's Use

(i) Identify the possible isomers of the alcohol, C_3H_8O.

[2]

(ii) The balanced equation of the reaction is
$3\ C_3H_8O + K_2Cr_2O_7 + 4\ H_2SO_4 \rightarrow 3\ C_3H_6O + Na_2SO_4 + Cr_2(SO_4)_3 + 7\ H_2O$

I State the colour change that the student would observe during the reaction.
from………………………..… to ………………………………

II Calculate the amount of $K_2Cr_2O_7$ used.

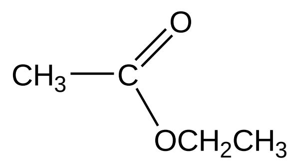
III The student obtained 5.22 g of the carbonyl compound, C_3H_6O. Calculate the percentage yield of the product obtained by the student assuming that the reaction goes to completion.

[5]

(iii) Use the table of characteristic values for Typical proton chemical shift values in the Data Booklet to answer this question.

For Examiner's Use

Nuclear magentic resonance can be used to identity functional groups in organic compounds. For example, ethyl ethanoate shows a chemical shift at 2.0ppm.



ethyl ethanoate

An impure sample of C_3H_6O obtained by a student was analysed using nuclear magnetic resonance. The nuclear magnetic resonance recorded a chemical shift at 9.0-13.0 ppm.

I Identity the functional group present in the impurity. Explain your answer.

Impurity ……………………………..

Reason…………………………………………………………………………

……………………………………………………………………………………..

……………………………………………………………………………………

II Suggest a structure for the alcohol used.

[4]

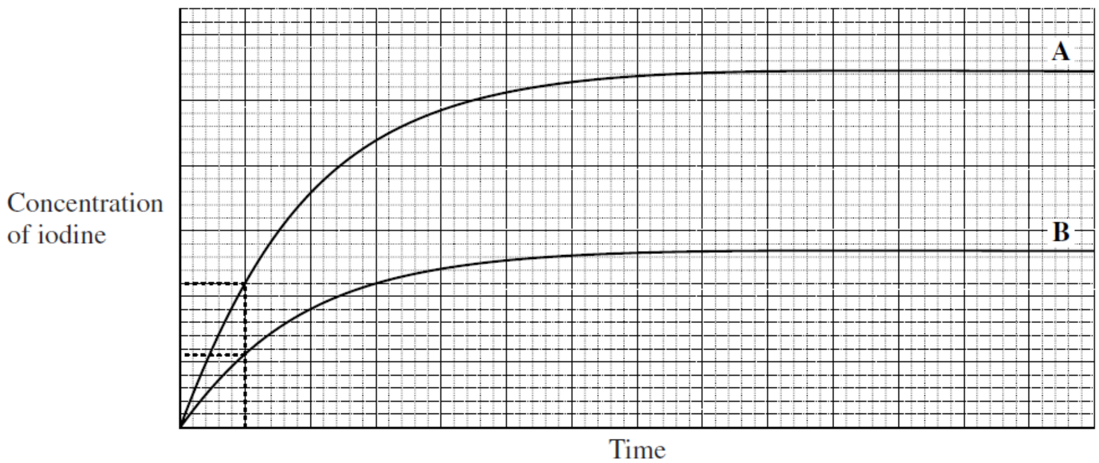
[Total:13]

For Examiner's Use

2 Iodide ions are oxidised to iodine by peroxodisulphate ions, $S_2O_8^{2-}$. The reaction can be catalysed by Fe^{2+}(aq) ions and by Fe^{3+}(aq) ions.

In an experiment to investigate the uncatalysed reaction, the concentration of iodine was determined at different times. Curve **A** shown below was obtained.

The experiment was repeated using half the original concentration of iodide ions but keeping other conditions the same. Curve **B** was obtained.



(a) Use these curves and the dotted lines to deduce the order of the reaction with respect to iodide ions. Explain how you deduced the order.

Order ……………………………………………………………………………………….

Explanation ………………………………………………………………………………...

…………………………………………………………………………………………....

……………………………………………………………………………………………

…………………………………………………………………………………………… [3]

(b) On the axes above, sketch a curve to show how the results will change if the experiment leading to curve **B** is repeated under the same conditions of concentration but at a lower temperature. Label this curve **X**. [2]

(c) On the axes above, sketch a curve to show how the results will change if the experiment leading to curve **A** is repeated in the presence of a catalyst containing Fe^{2+}(aq) ions. Label this curve **Y**. [2]

For Examiner's Use

(d) Suggest why the reaction needs to be catalysed by Fe^{2+}(aq) ions and by Fe^{3+}(aq) ions.

[2]

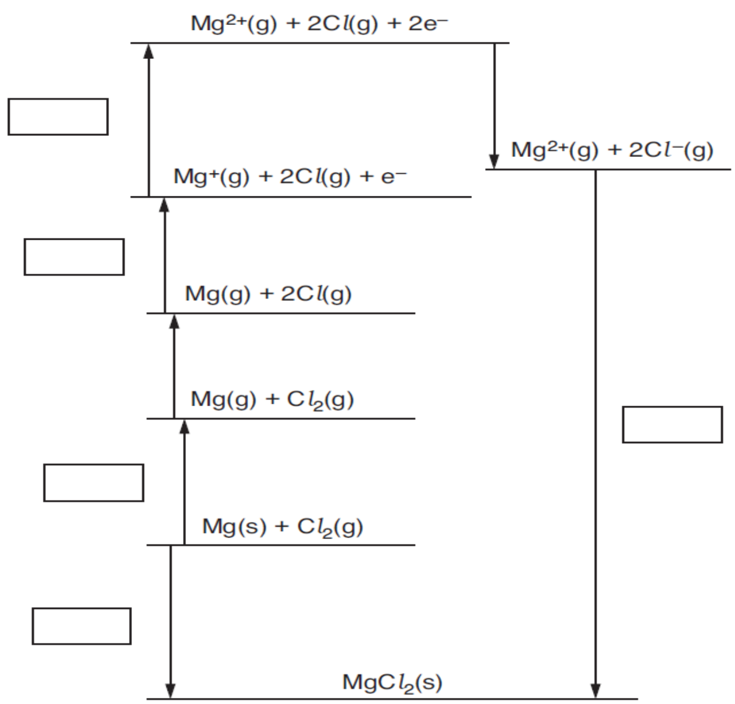
[Total:9]

3 Lattice enthalpy can be used as a measure of ionic bond strength. Lattice enthalpies are determined indirectly using an enthalpy cycle called a Born-Haber cycle.

The table below shows the enthalpy changes that are needed to determine the lattice enthalpy of magnesium chloride, $MgCl_2$.

Letter	Enthalpy change	Energy/ kJ mol^{-1}
A	1st electron affinity of chlorine	-349
B	1st ionization energy of magnesium	+736
C	2nd ionization energy of magnesium	+1450
D	Atomization of chlorine	+150
E	Atomistation of magnesium	+76
F	Formation of magnesium chloride	-642
G	Lattice enthalpy of magnesium chloride	

(a) On the cycle below, write the correct letter in each empty box.



[3]

For Examiner's Use

(b) Use the Born-Haber cycle to calculate the lattice enthalpy of magnesium chloride.

lattice enthalpy =……………… kJ mol^{-1} [2]

(c) Magnesium chloride melts at 714^{o}C while magnesium fluoride melts at 1255^{o}C Explain the difference in the melting points of the magnesium halides.

………………………………………………………………………………………………….

………………………………………………………………………………………………….

………………………………………………………………………………………………….

[2]

[Total:7]

4 A student obtained the following results when analysing an organic compound **H**.

Test		Observation
Test 1	Relative molecular mass	72
Test 2	% composition by mass	C, 66%; H, 11.1%; O, 22.2%
Test 3	Reactions with Br_2(aq)	Br_2 decolourised
Test 4	Reactions with Na(s)	H_2(g) evolved
Test 5	Reactions with warm $Cr_2O_7^{2-}/H^+$	Green colour observed

The student allowed test 5 to go to completion and then investigated the **product** of test 5 with the following result.

Test 6	Reaction with 2, 4-dinitrophenylhydrazine	No reaction

(a) Calculate the molecular formula of **H**.

[2]

(b) What can be deduced about the nature of **H** by the following tests?

(i) Test 3 ....

(ii) Test 4.. [2]

(c) **(i)** What functional group would have given a positive test result in test 6?

....

(ii) What functional group is shown to be present in **H** by tests 5 and 6?

.... [2]

(d) On testing a sample of **H**, the student found that it was not chiral.

H did, however, show *cis-trans* isomerism.

How does *cis-trans* isomerism arise in an organic molecule?

....

....

.... [2]

(e) Use all of the information above to draw *labelled, displayed formulae* of the stereoisomers of compound **H**.

[3]

[Total:11]

Section B: Free Response Questions [Total: 40 marks]

Answer **two** of three questions in this section on separate paper.

5 **(a)** **(i)** Calculate the oxidation number of carbon in methanol, CH_3OH.

(ii) Using **only** the elements C, H and O, draw the structural formulae of three compounds, **A, B** and **C**, each containing a single carbon atom with an oxidation number of zero, +2 and +4 respectively.

(iii) Suggest reagents and conditions for converting methanol into each of the three compounds **A, B** and **C.** [6]

(b) In an experiment to determine the enthalpy change of combustion of ethanol, ΔH_c, a quantity of the fuel was burned underneath a copper can containing 100g of water. It was found that the temperature of the water rose by 30.0°C after 1.50g of ethanol had been burned.

Assume that 50% of the heat obtained from burning the methanol is lost to the surroundings.

(i) Calculate the apparent ΔH_c of ethanol from these figures. Ignore the heat capacity of the copper can, and use the figure of 4.18 J g^{-1} K^{-1} for the specific heat capacity of water.

(ii) The true value of ΔH_c of ethanol is -1367 kJmol^{-1}. Compare this value to the one you have calculated in **(i)** and suggest a reason for the discrepancy. [4]

(c) Methanol can be produced by using a reversible reaction between carbon monoxide and hydrogen.

$$2H_2(g) + CO(g) \rightleftharpoons CH_3OH(g)$$

When 2.00 mol of hydrogen and 1.00 mol of carbon monoxide are mixed and heated to a high temperature in a container of volume 1.50 dm^3, the equilibrium yield of methanol is 0.80 mol.

(i) Write the equilibrium constant, K_c, for this reaction and give its units.

(ii) Calculate a value for the equilibrium constant, K_c, for this reaction at this temperature and give its units.

(iii) Taking the bond energy for the C-O bond in carbon monoxide to be 1077 kJ mol^{-1}, and using other appropriate bond energies given in the *Data Booklet*, calculated the entalpy change for this reaction.

Hence suggest why a pressure of 100 atm and 300°C are used in the reaction. [6]

(d) **(i)** Draw the structural formulae of four alcohols with the molecular formula $C_4H_{10}O$. Label your structures **D, E, F** and **G**.

(ii) Classify these alcohols as primary, secondary or tertiary.

(iii) Identify which alcohol reacts with alkaline aqueous iodine, giving the structural formulae of the products

(iv) Identify which alcohol is oxidised by an excess of acidified $K_2Cr_2O_7$ to give a non-acidic organic product, giving the structural formula of the product. [6]

[Total:20]

6 **(a)** Describe what is meant by the term *nucleon number*. [1]

(b) State **two** ways in which the behaviour of electrons in an electric field differs from that of protons. [1]

(c) **(i)** Draw dot-and-cross diagrams to show the bonding in the molecules of NO_2 and O_3.

(ii) Suggest a value for the bond angle in each of the above two molecules, giving reasons for your choice.

(iii) The compound FO_2 does not exist, but ClO_2 does.
By considering the possible types of bonding in the two compounds, suggest reasons for this difference. [6]

(d) **(i)** Describe how you would carry out the reaction between magnesium and oxygen, and state what you would observe during this reaction.

(ii) The pH values of the solutions formed when the chlorides of Period III are separately shaken with water decrease from sodium to phosphorus.

Explain why this is the case. [4]

(e) Ozone is usually made by passing oxygen gas through a tube between two highly charged electrical plates.

$$3O_2(g) \rightarrow 2O_3(g)$$

The reaction does not go to completion, so a mixture of the two gases results.
The concentration of O_3 in the mixture can be determined by its reaction with aqueous KI.

$$O_3 + 2KI + H_2O \rightarrow I_2 + O_2 + 2KOH$$

The iodine formed can be estimated by its reaction with sodium thiosulfate.

$$2Na_2S_2O_3 + I_2 \rightarrow Na_2S_4O_6 + 2NaI$$

When 500 cm^3 of an oxygen/ozone gaseous mixture at s.t.p. was passed into an excess of aqueous KI, and the iodine titrated, 15.0 cm^3 of 0.100 mol dm^{-3} $Na_2S_2O_3$ was required to discharge the iodine colour.

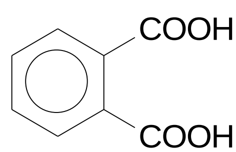
(i) Calculate the amount in moles of iodine produced.

(ii) Hence calculate the percentage of O_3 in the gaseous mixture. [3]

(f) Oxidation is an important reaction in organic chemistry. Both aldehydes and carboxylic acids can be prepared by the oxidation of alcohols with acidified $K_2Cr_2O_7$.

(i) Describe how you could ensure that only *either* the aldehyde **or** the carboxylic acid is produced during the oxidation process.

(ii) Compounds **L** and **M**, both C_9H_{10}, and compound **N**, $C_9H_{12}O$, are all oxidised by hot concentrated alkaline $KMnO_4$, followed by acidification, to give benzene-1, 2-dicarboxylic acid, **P**.



P

Compound **L** reacts with Br_2(aq), but compound **M** does not.

Compound **N** reacts with alkaline aqueous iodine.

Suggest structures for compounds **L, M** and **N**. [5]

[Total:20]

7 This question is about hydroxyacids.
One of the simplest hydroxyacids is lactic acid, 2-hydroxypropanoic acid. It can be made in the laboratory by the following route.

$$CH_3CHO \xrightarrow{\textbf{I}} CH_3CH(OH)CN \xrightarrow{\textbf{II}} \underset{\text{Lactic acid}}{CH_3CH(OH)COOH}$$

(a) State the reagents and conditions needed for
(i) Reaction I,
(ii) Reaction II.
[3]

The pH of 0.20 mol dm^{-3} lactic acid is 1.7.
(b) Explain whether lactic acid is a strong or weak acid with relevant calculation. [2]

(c) **(i)** Explain what is meant by a buffer solution.

Write equations that occur to illustrate the action of a buffer using a solution of lactic acid and sodium lactate as an example.

(ii) Suggest a suitable indicator for the titration of lactic acid with sodium hydroxide.

State the colour change at the end point. [4]

Poly(lactic acid) (PLA) is a biodegradable polymer used for food packaging and cosmetic bottles. A section of the PLA molecule is shown below.

$$-\!\left(OCH(CH_3)CO\right)_n\!-$$

PLA

(d) **(i)** Name the functional group present in PLA
(ii) Suggest the type of reaction that might occur during the biodegradation of PLA. [2]

(e) Another route to lactic acid acid uses the Hell-Volhard-Zelinsky (HVZ) reaction, followed by hydrolysis.

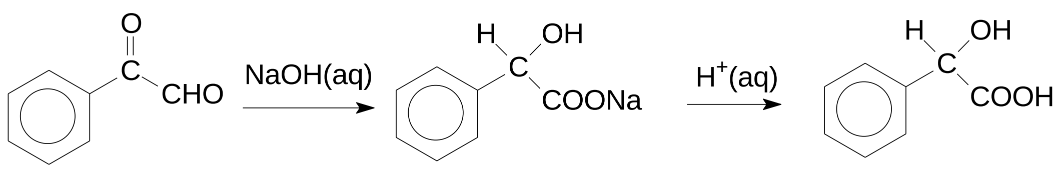
$$CH_3CH_2COOH \xrightarrow[\text{HVZ}]{Cl_2 + PCl_3} \underset{\textbf{J}}{CH_3CHClCOCl} \xrightarrow{\textbf{III}} CH_3CH(OH)COOH$$

(i) Suggest the reagents and conditions for reaction **III**.

(ii) When one mole of the dichloride **J** is treated with one mole of ethanol, C_2H_5OH, a neutral product **K** is obtained. Treatment of **K** with NH_3 in ethanol under pressure produces **L**, $C_5H_{11}NO_2$.

Suggest structures for **K** and **L** and explain the different reactivities of the two chlorine atoms in **J**. [5]

(f) 2-hydroxyphenylethanoic acid, **N**, can be prepared from compound **M** by the following route.



M **N**

Suggest **two** tests (stating reagents and observations) that would enable the compounds **M** and **N** to be distinguished from each other. [4]

[Total:20]